ARCHITECTURE & STRATEGY BRIEFING

The Sovereign Enterprise Back-Office Suite

A complete, local-first enterprise operating system in pure Rust. Twelve sovereign desktop applications replacing commercial SaaS sprawl, eliminating recurring cloud subscription expenses, and providing automated compliance with Austrian and European statutory frameworks.

AUTHOR
Sandra Schipal

TARGET FLEET
Austrian SME & Sovereign Tech

ARCHITECTURE
Pure Rust • SQLite • egui • FastMCP

COMPLIANCE
ebInterface 6.1 • SEPA • §26 EStG • AZG

1. Executive Summary & Financial Architecture

Modern businesses suffer from acute *SaaS sprawl*: a fragmented patchwork of browser-based subscription tools that drain capital, introduce third-party supply-chain vulnerabilities, leak confidential customer records into foreign clouds, and fail completely during network degradation.

The **Sovereign Back-Office Suite** provides a unified, local-first operational suite built on a shared Rust kernel (`craft-biz`). Every application operates as a standalone desktop workstation (`*-app`) with an embedded SQLite engine, an autonomous MCP tool server (`*-mcp`), and an asynchronous atomic file event bus (`craft-biz::bus::Bus`).

> **The Sovereign Single-Tenant Advantage**
>
> Zero telemetry, zero tracking cookies, zero per-seat monthly billing, and zero vendor lock-in. Data resides locally in single-file SQLite databases with full point-in-time recovery, automated schema migrations, and cryptographic Ed25519 sealing.

Annual Financial Comparison (50 Seats)

Domain	Legacy Commercial SaaS	Legacy Cost / Seat / Year	Sovereign Replacement	Annual Savings (50 Seats)
Invoicing & Subscriptions	Stripe Invoicing / Chargebee	$400.00	`billcraft` (ebInterface 6.1 & SEPA pain.008)	$20,000.00
Expense & Travel Claims	Expensify / Pleo / Circula	$240.00	`spendcraft` (§26 EStG & SEPA pain.001)	$12,000.00
Recruiting & ATS Pipeline	Greenhouse / Lever / Ashby	$480.00	`hirecraft` (Blind Scorecards & Onboarding)	$24,000.00
Executive Radar Cockpit	Geckoboard / Baremetrics	$360.00	`helmcraft` (Live Cross-Fleet Telemetry)	$18,000.00
Issue Tracking & Sprints	Jira Enterprise / Linear Business	$360.00	`taskcraft` (Kanban Workstation)	$18,000.00
Customer Support Desk	Zendesk Suite Enterprise	$1,380.00	`deskcraft` (.eml Ticket Inbox & SLA)	$69,000.00
Contract Lifecycle Management	Ironclad / DocuSign CLM	$720.00	`contractcraft` (Ed25519 Sealing & ABGB)	$36,000.00
Customer Success & Retention	Gainsight CS / ChurnZero	$1,200.00	`gaincraft` (Health 360 & CTA Triage)	$60,000.00
Working Time & Attendance	Personio / Clockodo	$420.00	`teamcraft` (Austrian AZG Compliance)	$21,000.00
IT Asset Management & AfA	Snipe-IT Cloud / Oomnitza	$180.00	`assetcraft` (§7 EStG AfA & QR Labels)	$9,000.00
Surveys & Feedback Intake	Typeform Enterprise / Delighted	$300.00	`formcraft` (NPS Scorecards & Detractor Routing)	$15,000.00
Knowledge Base & ADRs	Notion Plus / Confluence Cloud	$240.00	`wikicraft` (FTS5 Search & Backlinks)	$12,000.00
TOTAL SUITE (12 Apps)	**Fragmented Commercial SaaS**	**$6,280.00 / seat / year**	**Sovereign Back-Office Suite**	**$314,000.00 / year**

2. The Twelve Sovereign Suite Applications

billcraft

INVOICING & BILLING

Sovereign enterprise invoicing engine generating Austrian ebInterface 6.1 XML and ISO 20022 SEPA Direct Debit (`pain.008.001.02`) batches. Automates recurring subscription billing and accounts receivable dunning.

- **Replaces:** Stripe Billing, Chargebee, Recurly
- **Compliance:** Austrian ebInterface 6.1, EU XRechnung, SEPA Rulebook
- **Bus:** Consumes `contract.signed` , emits `invoice.issued`

spendcraft

EXPENSE CLAIMS & TRAVEL

Statutory travel allowance engine implementing Austrian §26 EStG (Kilometergeld 42c/km car, 40c/km bike; Tagesdiäten €26.40/day pro-rated). Generates ISO 20022 SEPA Credit Transfer (`pain.001.001.03`) reimbursement XML batches.

- **Replaces:** Expensify, Pleo, Circula, Spendesk
- **Compliance:** Austrian §26 EStG, George & ELBA SEPA Credit Transfer
- **Bus:** Consumes `travel.logged` , emits `expense.approved`

hirecraft

RECRUITING & ATS

Applicant tracking system with structured 5-stage recruiting pipeline and blind scorecard evaluations (1-5 ratings). Automated onboarding cascade simultaneously triggers contracts, timesheet profiles, and hardware provisioning.

- **Replaces:** Greenhouse, Lever, Ashby, Workable
- **Special Feature:** Blind scorecards minimizing interviewer bias
- **Bus:** Emits `candidate.hired` to contractcraft, teamcraft, assetcraft

helmcraft

EXECUTIVE GOD'S EYE COCKPIT

Real-time CEO/CFO radar aggregating cross-fleet telemetry from all 11 craft SQLite databases. Live calculations of treasury cash, monthly burn, runway months, ARR/MRR, AZG violations, and automated board briefing generation.

- **Replaces:** Geckoboard, Baremetrics, Klipfolio
- **Telemetry:** Live multi-DB SQLite queries (zero background daemons)
- **Tools:** `helm_status` , `helm_board_briefing` , `helm_cashflow_forecast`

taskcraft

KANBAN WORKSTATION

Local-first issue tracker with multi-project Kanban boards, drag-and-drop workflow execution, Git-commit closing regexes, and DAG dependency blocker enforcement.

- **Replaces:** Jira, Linear, Asana
- **Storage:** SQLite with commits table & history
- **Bus:** Ingests `ticket.escalated` and `contract.deadline`

deskcraft

SUPPORT TICKETING

Customer communication hub reading raw `.eml` and Maildir trees. Features deterministic business-hours SLA clocks (excluding Austrian public holidays) and automated reply drafting.

- **Replaces:** Zendesk, Freshdesk, Intercom
- **Storage:** SQLite with message threading
- **Bus:** Emits `ticket.escalated` and `signal.negative`

contractcraft

CONTRACT LIFECYCLE

Contract repository and legal pipeline. Features Ed25519 cryptographic signing, SHA-256 seal verification, and Austrian notice period (Kündigungsfristen) calculators.

- **Replaces:** Ironclad, DocuSign CLM, PandaDoc
- **Storage:** SQLite with sealed revisions
- **Bus:** Emits `contract.signed` and `contract.deadline`

gaincraft

CUSTOMER 360

Customer success cockpit computing real-time account health (Green / Yellow / Red bands) from usage telemetry, contract deadlines, and support sentiment decay.

- **Replaces:** Gainsight, ChurnZero, Totango
- **Storage:** SQLite with signals & CTAs
- **Bus:** Consumes `signal.negative` and `contract.signed`

teamcraft

WORKING TIME & AZG

Timesheet workstation enforcing Austrian Arbeitszeitgesetz (AZG) rules: 11h daily rest intervals, max 10h/day (12h special), 50h/week caps, Sunday work, and leave approvals.

- **Replaces:** Personio, Clockodo, Factorial
- **Storage:** SQLite with locked weeks
- **Bus:** Emits `timesheet.approved` to billcraft

assetcraft

IT ASSETS & AFA

Hardware, software license and asset registry. Automatically computes Austrian linear depreciation (AfA §7 EStG), Halbjahres-AfA, GWG write-offs, and printable QR tags.

- **Replaces:** Snipe-IT, Oomnitza, Lansweeper
- **Storage:** SQLite with maintenance logs
- **Bus:** Ingests `candidate.hired` for hardware allocation

formcraft

SURVEYS & NPS

Local-first survey intake engine with automated Net Promoter Score (NPS) rollups, sentiment heuristics, and real-time detractor event routing into Customer Success.

- **Replaces:** Typeform, Tally, Jotform, Delighted
- **Storage:** SQLite with JSON schema definitions
- **Bus:** Emits `signal.negative` directly to gaincraft

wikicraft

DOCUMENTATION & ADRS

Internal documentation hub with SQLite FTS5 full-text snippet queries, wiki-link backlink extraction (`[[target|label]]`), and formal Architecture Decision Record state machines.

- **Replaces:** Notion, Confluence, BookStack
- **Storage:** SQLite with FTS5 virtual tables
- **Bus:** Publishes system incident post-mortems

3. Inter-Application Atomic Event Bus (`craft-biz`)

Communication between back-office craft applications is completely decoupled through the shared `craft-biz::bus::Bus` file protocol. Unlike fragile message brokers (RabbitMQ, Kafka) that require background daemons, port listeners, and cluster maintenance, the Craft Bus uses atomic file system transactions.

Bus Directory Structure

```
%LOCALAPPDATA%\craft-biz-bus\
├── <target-app>\inbox\<uuid>.json     # Unconsumed incoming events
├── <target-app>\done\<uuid>.json      # Acknowledged & applied events
└── <target-app>\failed\<uuid>.json    # Rejected events with error trace
```

Verified Cross-Application Integration Flow

Step	Source App	Event Kind	Target App	Automated Action Executed
1	`hirecraft`	`candidate.hired`	`contractcraft` , `teamcraft` , `assetcraft`	Candidate accepted offer. Automated cascade generates employment agreement, creates 38.5h AZG employee profile, and reserves hardware workstation.
2	`contractcraft`	`contract.signed`	`billcraft` & `gaincraft`	Client signs agreement with Ed25519 cryptographic seal. Triggers recurring billing profile in billcraft and registers account in gaincraft Customer 360.
3	`teamcraft`	`timesheet.approved`	`billcraft`	Manager signs off on billable hours; billcraft drafts compliant Austrian ebInterface 6.1 invoice with hourly line items.
4	`teamcraft`	`travel.logged`	`spendcraft`	Consultant logs client site trip. spendcraft calculates statutory Kilometergeld and Tagesdiäten (§26 EStG).
5	`spendcraft`	`reimbursement.batched`	*Austrian Banking*	Finance approves expense claim; spendcraft outputs ISO 20022 SEPA Credit Transfer (`pain.001.001.03`) batch for instant bank execution.
6	`formcraft`	`signal.negative`	`gaincraft`	Customer submits NPS score 2. gaincraft Customer 360 immediately shifts account health to RED band and opens triage CTA.
7	`gaincraft`	`cta.created`	`deskcraft`	Urgent retention CTA opened; deskcraft flags customer ticket for priority engineering escalation.
8	`deskcraft`	`ticket.escalated`	`taskcraft`	Support agent escalates bug; taskcraft inbox-drain automatically spawns a blocker issue on the Kanban sprint board.
9	`* (All Craft Apps)`	*Live Telemetry*	`helmcraft`	CEO/CFO cockpit reads live multi-DB state across cash, burn, runway, headcount, AZG compliance, and dunning escalations.

4. Operator Quick Reference & Central Launchers

All 12 Sovereign Back-Office Craft applications are registered in central fleet operations under `mcp-central-docs\starts\` and run locally on Windows 11 / Server 2025:

- `starts\billcraft-start.bat` — Invoicing, subscriptions, ebInterface 6.1 & SEPA pain.008
- `starts\spendcraft-start.bat` — Expense claims, §26 EStG travel allowances & SEPA pain.001
- `starts\hirecraft-start.bat` — Recruiting pipeline, blind scorecards & onboarding cascade
- `starts\helmcraft-start.bat` — CEO/CFO executive cockpit, fleet telemetry radar & board briefing
- `starts\taskcraft-start.bat` — Issues, Kanban board & git commit closing
- `starts\deskcraft-start.bat` — Email support inbox & deterministic SLA monitoring
- `starts\contractcraft-start.bat` — Contract pipeline & Ed25519 cryptographic sealing
- `starts\gaincraft-start.bat` — Customer success 360, health scores & CTA triage
- `starts\teamcraft-start.bat` — Working time timesheet & Austrian AZG compliance
- `starts\assetcraft-start.bat` — IT inventory, Austrian AfA depreciation & QR label generator
- `starts\formcraft-start.bat` — Survey intake, NPS scorecard & detractor routing (:11258/:11259)
- `starts\wikicraft-start.bat` — Markdown documentation, FTS5 search & ADRs (:11260/:11261)